kn Guide

> Applies to macOS 26 / Keynote 15.4 (v14_5 schema, compatible with M6.5—M15.4)
>
> Status: Rust 3.0 — all 57 commands ported
>
> Principle: normal commands read the source and write a new file; commands that need Keynote's document model briefly launch Keynote

kn reads and edits Apple Keynote .key files. Most operations work directly on the IWA / Snappy / protobuf container. new, add-slide, change-layout, insert-image, insert-image-paste, del-slide, dup-slide, and the Keynote-backed table fill/merge paths use Keynote itself.

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1. Install and run

```
cd /path/to/project
./bin/kn help
./bin/kn outline deck.key
```

./bin/kn builds target/release/kn on first use when the binary is missing.

Manual build:

```
cargo build --release
./bin/kn verify deck.key
```

Requirements:

- Rust 1.80+ (edition 2021)
- protoc for Keynote protobuf code generation
- macOS Keynote for the Keynote-backed commands listed above

No external runtime schema package is required.

Exit codes

Code	Meaning
0	Success
1	Internal error / unsupported structure / write-back validation failed
2	Usage error, missing file, page out of range, invalid argument

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2. Five-minute quick start

```
# Inspect
./bin/kn outline deck.key
./bin/kn texts   deck.key
./bin/kn verify  deck.key

# Title and body
./bin/kn set-text in.key out.key 3 title "New title"
./bin/kn set-text in.key out.key 3 body  "New body"

# Text color and style
./bin/kn set-text-color in.key out.key 3 title "#FF00FF"
./bin/kn set-text-style in.key out.key 3 title --size 48 --bold

# Table cell
./bin/kn tables deck.key
./bin/kn set-cell in.key out.key 7031011 2 2 777

# Final check
./bin/kn audit out.key
```

Pages and table rows/columns are 1-based. Shape paths and merge-cells coordinates are 0-based; see the relevant sections.

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3. Command overview

Area	Count	Commands
Inspect	17	list verify text texts types outline tables charts images links builds builds-list transitions styles colors audit check-notes
Text / notes	6	set-text set-text-color set-text-style note set-note edit
Color / shape	9	set-bg set-shape set-color set-style set-shape-geometry group ungroup del-shape dup-shape
Table	9	set-cell add-row del-row add-col del-col set-table-border set-table-fill set-table-size merge-cells
Chart / image / link / animation	9	set-chart set-image image-info insert-image insert-image-paste set-link add-build set-build set-transition
Page structure	5	add-slide change-layout del-slide dup-slide skip
Batch	1	batch

Area	Count	Commands
Create	1	new

info is an alias of verify; skip-many is another entry point for skip. del-slide accepts page expressions such as 3, 1-3,7, and all.

Write commands use <src> <dst> and do not overwrite the source. For commands that take an ident, query it first with the matching inspection command.

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4. Inspect

```
./bin/kn list       deck.key
./bin/kn verify     deck.key
./bin/kn text       deck.key [keyword]
./bin/kn texts      deck.key [keyword]
./bin/kn types      deck.key
./bin/kn outline    deck.key
./bin/kn tables     deck.key
./bin/kn charts     deck.key
./bin/kn images     deck.key
./bin/kn links      deck.key
./bin/kn builds     deck.key
./bin/kn builds-list deck.key
./bin/kn transitions deck.key
./bin/kn styles     deck.key [keyword]
./bin/kn colors     deck.key
./bin/kn audit      deck.key
./bin/kn check-notes deck.key [--require text]
```

Notes:

- text searches raw strings; texts reads TSWP.StorageArchive text objects.
- colors lists inline RGB color instances. Its index is the set-color selector.
- check-notes defaults to requiring 内容依据 and 设计依据; --require sets one required substring.
- audit performs lightweight structure, identifier, page, and note checks.

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5. Text and notes

```
# Whole title/body
./bin/kn set-text in.key out.key 3 title "New title"
./bin/kn set-text in.key out.key 3 body  "New body"

# Size, bold, italic, color; optionally a substring
```

```
./bin/kn set-text-style in.key out.key 3 title --size 48 --bold
./bin/kn set-text-style in.key out.key 3 title --italic --color "#C82505"
./bin/kn set-text-style in.key out.key 3 title --substr "keyword" --color "#FF0000"

# Whole-run text color
./bin/kn set-text-color in.key out.key 3 title "#FF00FF"
./bin/kn set-text-color in.key out.key 3 body  "#00A000"

# Notes: append or replace the whole note
./bin/kn note     in.key out.key 3 "appended note"
./bin/kn set-note in.key out.key 3 "replacement note"

# Raw replacement; use "-" to search all .iwa files
./bin/kn edit in.key out.key - "old" "new" --resize
```

- title / body select the page placeholders.
- set-text-color is whole-run only; it has no --substr / --range flags.
- set-note replaces the whole note. Use note when existing text must be kept.
- edit without --resize requires equal byte length; --resize recomputes nested protobuf lengths.

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6. Color, styles, and shapes

```
./bin/kn set-bg    in.key out.key 3 "#EEEEEE"
./bin/kn set-shape in.key out.key 3 2 "#C82505"
./bin/kn set-color in.key out.key 1943 "#FF00FF" --alpha 0.6
./bin/kn set-style in.key out.key 7031016 font_size=48
./bin/kn set-style in.key out.key 7031016 bold=true

./bin/kn set-shape-geometry in.key out.key 3 2 x=100 y=200 w=600 h=120 angle=15
./bin/kn dup-shape in.key out.key 3 3 --offset 40,40
./bin/kn del-shape in.key out.key 3 2

./bin/kn group   in.key out.key 3 "2,3"
./bin/kn ungroup in.key out.key 3 2
```

- **Shape paths are 0-based**: 2 is the third top-level object; 3/1 is the second child of the fourth top-level group.
- set-shape only handles a solid fill. Gradient or image fills produce an error.
- dup-shape supports images and ordinary shapes; use table commands for tables and duplicate a page for placeholders.
- set-style changes one property at a time: font_size=48, bold=true, italic=false, font_color=#FF0000.

- Do not group title/body placeholders: grouping removes their placeholder identity. Confirm object paths with outline first.

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7. Tables

Find the table ident first:

```
./bin/kn tables deck.key
```

Cells and rows/columns

```
# set-cell / add-row / del-row / add-col / del-col: 1-based rows and columns
./bin/kn set-cell in.key out.key 7031011 2 2 777
./bin/kn add-row  in.key out.key 7031011 1 --at 2
./bin/kn del-row  in.key out.key 7031011 3
./bin/kn add-col  in.key out.key 7031011 --at 3
./bin/kn del-col  in.key out.key 7031011 3

# Row height / column width
./bin/kn set-table-size in.key out.key 7031011 --row-height 80 --col-width 180
```

del-col is experimental because column structure is coupled to several tables; always reopen the output in Keynote.

Borders, fills, merges

```
./bin/kn set-table-border in.key out.key 7031011 color="#EE2222" width=4
./bin/kn set-table-fill   in.key out.key 7031011 --body "#FF00FF" --no-band
./bin/kn set-table-fill   in.key out.key 7031011 --header "#EEF2F6" --header-rows 1
./bin/kn set-table-fill   in.key out.key 7031011 --band "#F2F2F7"

# merge-cells: 0-based rows/columns, top-left to bottom-right
./bin/kn merge-cells in.key out.key 7031011 0 0 1 1
```

- set-table-fill sets cell background colors through Keynote and needs Automation permission.
- set-table-border writes a table-style variation; reopen the output in Keynote.
- merge-cells coordinates are 0-based, unlike the 1-based set-cell.

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8. Charts, images, links, animation, transitions

```
# Charts: 0-based rows/columns
./bin/kn charts    deck.key
./bin/kn set-chart in.key out.key 7031018 0 0 100
```

```
# Images
./bin/kn images deck.key
./bin/kn image-info deck.key 7034153
./bin/kn set-image in.key out.key 7031019 new.png
./bin/kn insert-image in.key out.key 3 new.png
./bin/kn insert-image-paste in.key out.key 3 new.png

# Links
./bin/kn links    deck.key
./bin/kn set-link in.key out.key 7031020 "https://example.com"

# Animation and transition
./bin/kn builds-list deck.key
./bin/kn add-build   in.key out.key 3 2
./bin/kn set-build   in.key out.key 7035366 duration=2.5 automatic=true
./bin/kn transitions deck.key
./bin/kn set-transition in.key out.key 3 apple:dissolve 1.5
```

- insert-image and insert-image-paste are performed by Keynote. The paste path needs Accessibility and Keynote Automation permissions.
- set-image replaces the Data/ payload and updates natural/original dimensions.
- image-info reports one image's data byte count and encoded dimensions; use it to confirm the target ident before replacing.
- add-build uses a 0-based drawable path and requires an existing build template in the document.

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9. Pages and creation

```
# Add a slide (generated by Keynote)
./bin/kn add-slide in.key out.key "Title & Bullets" \
  --title "Title" --body "Body" --after 3

# Change layout / delete / duplicate
./bin/kn change-layout in.key out.key 5 "Title & Bullets"
./bin/kn del-slide in.key out.key "3,7-9"
./bin/kn dup-slide in.key out.key 5

# Skip slides: 3 / 1-3,7 / all
./bin/kn skip in.key out.key "2,4"
./bin/kn skip in.key out.key "2,4" --off
```

add-slide --replace N replaces slide N with the target layout and migrates its title/body. The layout name must exist in the current theme.

Create from scratch

```
./bin/kn new new.key --title "Title" --body "First point"
./bin/kn new new.key --theme "Basic White" --layout "Title & Bullets"
./bin/kn new new.key --force
```

The defaults are the built-in Basic White theme and Title & Bullets layout. Keynote creates the document and closes it after saving. Existing files are not overwritten unless --force is given.

Batch

```
./bin/kn batch in-dir out-dir set-text 1 title "Unified title"
./bin/kn batch in-dir out-dir set-text-style 1 title --color "#C82505"
./bin/kn batch in-dir out-dir skip 3
```

Each .key in the input directory runs the same subcommand; results keep their file names in the output directory.

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10. Common workflows

Batch title and color

```
for f in *.key; do
  ./bin/kn set-text "$f" "out/$f" 1 title "Course title"
  ./bin/kn set-text-color "out/$f" "out2/$f" 1 title "#1D4ED8"
done
```

Table cell

```
./bin/kn tables report.key
./bin/kn set-cell report.key out.key 7031011 2 2 777
```

Replace an image

```
./bin/kn images gallery.key
./bin/kn set-image gallery.key out.key 7034303 new.jpg
```

Add an entrance build

```
./bin/kn builds-list animation.key
./bin/kn add-build animation.key out.key 2 0 apple:bc-appear
```

Table fill and merge

```
./bin/kn set-table-fill report.key out.key 7031011 \
  --header "#1F3864" --body "#FFFFFF" --header-rows 1
./bin/kn merge-cells out.key out2.key 7031011 0 0 1 1
```

> Find the table ident with tables; after writing, run verify and open the output in Keynote. Keep the intermediate copy.

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11. Verification and troubleshooting

```
./bin/kn verify deck.key
./bin/kn audit deck.key
./bin/kn check-notes deck.key --require "内容依据"
```

A structurally valid file is not necessarily rendered correctly. Always open the output in Keynote before delivery; for colors, images, and table styles, use Keynote-exported PNG/PDF page counts and pixels as the final check.

Symptom	Likely cause	Fix
Output will not open	Unknown structure touched	Run verify, revert to last good version, quit and reopen Keynote
insert-image-paste fails	Missing Accessibility/Automation permission	Grant permission to the terminal or DSH
add-slide / new fails	Keynote missing, permission denied, theme/layout not found	Open Keynote and check theme/layout names
Encrypted file error	Password-protected document	Remove the password in Keynote and save a copy
iWork '09 rejected	Legacy index.apxl format	Save as a current Keynote document
Output already exists	new does not overwrite by default	Use --force or another output path

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12. Safety boundaries

- **Source is read-only**: write commands use <src> <dst>; keep the original.

- **Keynote paths**: new, add-slide, change-layout, insert-image, insert-image-paste, del-slide, dup-slide, table fill/merge briefly open Keynote.
- **Experimental**: del-col, set-table-border, set-table-fill involve structural/style rewrites; verify in Keynote. set-shape only supports solid fills.
- **Version scope**: mainly validated on macOS 26 / Keynote 15.4; older versions were not tested exhaustively.
- **Concurrency/undo**: simultaneous edits and Keynote's per-slide undo behavior were not specifically validated.
- **Distribution**: the public repository does not include real .key samples or personal paths.

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13. Format report

docs/AppleKeynoteFormatReport.pdf is a 24-page analysis of the .key format. Example file names are generic.

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